6 Databases (109)

6.0.1 NIELIT 2018-51

_____ is holding an entry for each terminal symbol and is acting as permanent database.

A. Variable Table
B. Terminal Table
C. Keyword Table
D. Identifier Table

nielit-2018 databases

Answer key☟

6.0.2 NIELIT Scientist B 2020 November: 110

An attribute(s) that is used to look up for records in a file is called a:

A. Function key
B. Catalog key
C. Access key
D. Search key

nielit-scb-2020 databases

Answer key☟

6.0.3 NIELIT Scientist B 2020 November: 107

Most NoSQL databases support automatic ___________ meaning that you get high availability and disaster recovery.

A. Processing
B. Scalability
C. Replication
D. All of the options

nielit-scb-2020 databases

Answer key☟

6.1 Aggregate Functions (1)

6.1.1 Aggregate Functions: NIELIT 2022 April Scientist B | Section B | Question: 115

The employee information in a company is stored in the relation

Employee (name, sex, salary, deptName)

Consider the following SQL query

```
Select deptName
  From Employee
  Where sex = 'M'
  Group by deptName
  Having avg(salary) >
    (select avg (salary) from Employee)
```

It returns the names of the department in which

A. the average salary is more than the average salary in the company
B. the average salary of male employees is more than the average salary of all male

employees in the company

C. the average salary of male employees is more than the average salary of employees in same the department

D. the average salary of male employees is more than the average salary in the company

nielit2022apr-scientistb sql databases query aggregate-functions

6.2 B Tree (3)

6.2.1 B Tree: NIELIT 2017 DEC Scientific Assistant A - Section B: 21

The $2-3-4$ tree is a self-balancing data structure, which is also called :

A. $2-4$ tree
B. $B+$ tree
C. $B-$ tree
D. None of the options

nielit2017dec-assistanta databases b-tree

Answer key☟

6.2.2 B Tree: NIELIT 2017 July Scientist B (CS) - Section B: 43

Which one of the following is a key factor for preferring B-trees to binary search trees for indexing database relations?

A. Database relations have a large number of records
B. Database relations are sorted on the primary key
C. B-trees require less memory than binary search trees
D. Data transfer from disks is in blocks

nielit2017july-scientistb-cs databases b-tree

Answer key☟

6.2.3 B Tree: NIELIT 2021 Dec Scientist A - Section B: 58

Non leaf nodes of $\mathrm{B}+$ tree structure from a :

A. Multilevel sparse indices
B. Multilevel dense indices
C. Sparse indices
D. Multilevel clustered indices

nielit2021dec-scientista data-structures b-tree databases

Answer key☟

6.3 Bcnf (2)

6.3.1 Bcnf: NIELIT 2017 DEC Scientist B - Section B: 24

Which of the following is TRUE?

A. Every relation in 3NF is also in BCNF
B. A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent

on every key of R

C. Every relation in BCNF is also in 3NF

D. No relation can be in both BCNF and 3NF.

nielit2017dec-scientistb database-normalization bcnf

Answer key☟

6.3.2 Bcnf: NIELIT 2017 DEC Scientist B - Section B: 55

Consider the relational schema $\mathrm{R(A\ B\ C\ D)}$ with following functional dependency set $F = \{A \rightarrow BC, C \rightarrow D\}$; The relation R is in

A. 2NF B. BCNF C. 3NF D. 1NF

nielit2017dec-scientistb database-normalization bcnf

Answer key☟

6.4 Bcnf Decomposition (2)

6.4.1 Bcnf Decomposition: NIELIT 2016 DEC Scientist B (IT) - Section B: 44

Which one of the following statements about normal forms is FALSE?

A. BCNF is stricter than 3NF.

B. Lossless, dependency-preserving decomposition into BCNF is always possible.

C. Lossless, dependency-preserving decomposition into 3NF is always possible.

D. Any relation with two attributes is BCNF.

nielit2016dec-scientistb-it databases database-normalization bcnf-decomposition

Answer key☟

6.4.2 Bcnf Decomposition: NIELIT 2017 DEC Scientist B - Section B: 1

Which one of the following statements are not correct?

$S1$: 3NF decomposition is always lossless join and dependency preserving.

$S2$: 3NF decomposition is always lossless join but may or may not be dependency preserving.

$S3$: BCNF decomposition is always lossless join and dependency preserving.

$S4$: BCNF decomposition is always lossless join but may or may not be dependency preserving.

A. Only $S1$ B. Only $S4$ C. Both $S1$ and $S4$ D. Both $S2$ and $S3$

nielit2017dec-scientistb database-normalization bcnf-decomposition

Answer key☟

6.5 Btrees (1)

6.5.1 Btrees: NIELIT 2018-34

Identify the true statement from the given statements:

1. Number of child pointers in a B/B+ tree node is always equal to number of keys in it plus one
2. B/B+ tree is defined by a term minimum degree. And minimum degree depends on hard disk block size, key and address size

A. (1) B. (1) and (2) C. (2) D. None of these

nielit-2018 databases btrees

Answer key☟

6.6 Candidate Key (5)

6.6.1 Candidate Key: NIELIT 2016 DEC Scientist B (CS) - Section B: 54

If there is more than one key for relation schema in DBMS then each key in relation schema is classified as

A. prime key B. super key C. candidate key D. primary key

nielit2016dec-scientistb-cs databases candidate-key

Answer key☟

6.6.2 Candidate Key: NIELIT 2016 DEC Scientist B (IT) - Section B: 36

The primary key is selected from the:

A. Composite keys B. Determinants
C. Candidate keys D. Foreign keys

nielit2016dec-scientistb-it databases candidate-key

Answer key☟

6.6.3 Candidate Key: NIELIT 2016 MAR Scientist B - Section C: 43

L e t $R = (A, B, C, D, E, F)$ be a relation scheme with the following dependencies:

$C \to F, E \to A, EC \to D, A \to B.$

Which of the following is a key for R?

A. CD B. EC C. AE D. AC

nielit2016mar-scientistb database-normalization candidate-key

Answer key☟

6.6.4 Candidate Key: NIELIT Scientific Assistant A 2020 November: 72

Let $\mathrm{R}{=}(\mathrm{A,B,C,D,E})$ having following FDs. $\mathrm{F} = \{\mathrm{A} \rightarrow \mathrm{BC}, \mathrm{CD} \rightarrow \mathrm{E}, \mathrm{B} \rightarrow \mathrm{D}, \mathrm{E} \rightarrow \mathrm{A}\}$.

Which of the following is not a Candidate key?

A. A B. B C. E D. BC

nielit-sta-2020 databases easy database-normalization candidate-key

Answer key☟

6.6.5 Candidate Key: NIELIT Scientist B 2020 November: 96

An instance of relational schema $R(A, B, C)$ has distinct values of A including $NULL$ values. Which one of the following is true?

A. A is a candidate key
B. A is not a candidate key
C. A is a primary key
D. Both “A is a candidate key” and “A is a primary key”

nielit-scb-2020 relational-algebra databases candidate-key

Answer key☟

6.7 Data Dependency (4)

6.7.1 Data Dependency: NIELIT 2016 MAR Scientist C - Section C: 34

Given the following relation instance:

X	Y	Z
1	4	2
1	5	3
1	6	3
3	2	2

Which of the following functional dependencies are satisfied by the instance?

A. $XY \rightarrow Z$ and $Z \rightarrow Y$
B. $YZ \rightarrow X$ and $Y \rightarrow Z$
C. $YZ \rightarrow X$ and $X \rightarrow Z$
D. $XZ \rightarrow Y$ and $Y \rightarrow X$

nielit2016mar-scientistc databases data-dependency

Answer key☟

6.7.2 Data Dependency: NIELIT 2016 MAR Scientist C - Section C: 35

Consider the schema $R = (\mathrm{S\ T\ U\ V})$ and the dependencies $S \rightarrow T, T \rightarrow U, U \rightarrow V$, and $V \rightarrow S$. If $R = (R1 \text{ and } R2)$ be a decomposition such that $R1 \cap R2 = \phi$ then decomposition is

A. not in 2 NF
B. in 2 NF but not in 3 NF
C. in 3 NF but not in 2 NF
D. in both 2 NF and 3 NF

nielit2016mar-scientistc databases data-dependency

Answer key☟

6.7.3 Data Dependency: NIELIT 2016 MAR Scientist C - Section C: 4

A functional dependency of the form $x \rightarrow y$ is trivial if

A. $y \subseteq x$
B. $y \subset x$
C. $x \subseteq y$
D. $x \subset y$ and $y \subset x$

nielit2016mar-scientistc databases data-dependency

Answer key☟

6.7.4 Data Dependency: NIELIT 2018-58

Consider the following functional dependencies in a database:

$A \rightarrow B$

$B \rightarrow C$

$D \rightarrow E$

$E \rightarrow D$

$F \rightarrow G$

$F \rightarrow H$

$(E, F) \rightarrow I$

The relation (E, D, A, B) is

A. 2 NF
B. 3 NF
C. BCNF
D. None of the above

nielit-2018 databases data-dependency

Answer key☟

6.8 Database Design (2)

6.8.1 Database Design: NIELIT 2021 Dec Scientist A - Section B: 57

Consider a relation R with attributes $\{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ and functional dependency set $\mathrm{S}=\{\mathrm{A} \rightarrow \mathrm{B}, \mathrm{A} \rightarrow \mathrm{C}\}$. Then relation R can be decomposed into two relations :

A. $\mathrm{R1}\{\mathrm{A}, \mathrm{B}\}$ AND $\mathrm{R2}\{\mathrm{A}, \mathrm{C}\}$
B. $\mathrm{R1}\{\mathrm{A}, \mathrm{B}\}$ AND $\mathrm{R2}\{\mathrm{B}, \mathrm{C}\}$
C. $\mathrm{R1}\{\mathrm{A}, \mathrm{B}, \mathrm{C}\}$ AND $\mathrm{R2}\{\mathrm{A}, \mathrm{C}\}$
D. None of the above

nielit2021dec-scientista databases database-design normal-forms functional-dependency relational-model

Answer key☟

6.8.2 Database Design: NIELIT 2021 Dec Scientist B - Section B: 65

Assume that a DBA issued the following create table command :

create table A $(\mathrm{Aid}, \ldots\ldots)$

storage $(\text{initial } 20480, \text{next } 20480, \text{maxextents } 8, \text{minextents } 3, \text{pctincrease } 0);$

How many bytes of disk space will be allocated to this file when it is first created ?

A. $163,840$ bytes
B. 20480 bytes
C. $61,440$ bytes
D. 8 bytes

nielit2021dec-scientistb databases disk database-design

Answer key☟

6.9 Database Normalization (11)

6.9.1 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 2

Process of analyzing relation schemas to achieve minimal redundancy and insertion or update anomalies is classified as:

A. normalization of data.
B. denomination of data.
C. isolation of data.
D. de-normalization of data.

nielit2016dec-scientistb-cs databases database-normalization

Answer key☟

6.9.2 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 31

Rule which states that addition of same attributes to right side and left side will result in other valid dependency is classified as

A. referential rule
B. inferential rule
C. augmentation rule
D. reflexive rule

nielit2016dec-scientistb-cs databases database-normalization

Answer key☟

6.9.3 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 47

If every functional dependency in set E is also in closure of F then this is classified as

A. FD is covered by E
B. E is covered by F
C. F is covered by E
D. F plus is covered by E

nielit2016dec-scientistb-cs databases database-normalization

Answer key☟

6.9.4 Database Normalization: NIELIT 2016 DEC Scientist B (CS) - Section B: 49

Considering relational database, functional dependency between two attributes A and B is denoted by

A. $A \rightarrow B$ B. $B \leftarrow A$ C. $AB \rightarrow R$ D. $R \leftarrow AB$

nielit2016dec-scientistb-cs databases database-normalization

Answer key

6.9.5 Database Normalization: NIELIT 2016 DEC Scientist B (IT) - Section B: 14

Normalization from which is based on transitive dependency is classified as:

A. First normal form.
B. Second normal form.
C. Fourth normal form.
D. Third normal form.

nielit2016dec-scientistb-it databases database-normalization

Answer key

6.9.6 Database Normalization: NIELIT 2016 DEC Scientist B (IT) - Section B: 40

In functional dependency between two sets of attribute A and B then set of attributes A of database is classified as:

A. top right side B. down left side C. left hand side D. right hand side

nielit2016dec-scientistb-it databases database-normalization

Answer key

6.9.7 Database Normalization: NIELIT 2016 DEC Scientist B (IT) - Section B: 57

If attribute of relation schema R is member of some candidate key then this type of attributes are classified as:

A. atomic attribute
B. candidate attribute
C. non-prime attribute
D. prime attribute

nielit2016dec-scientistb-it databases database-normalization

Answer key

6.9.8 Database Normalization: NIELIT 2016 MAR Scientist B - Section C: 44

For a database relation $R(a, b, c, d)$ where the domains of a, b, c, d include only atomic values, only the following functional dependencies and those that can be inferred from them hold:

$a \rightarrow c,$

$b \rightarrow d$

the relation is in

A. first normal form but not in second normal form.
B. second normal form but not in third normal form.
C. third normal form.
D. none of these.

nielit2016mar-scientistb databases database-normalization

Answer key

6.9.9 Database Normalization: NIELIT 2017 DEC Scientist B - Section B: 25

Consider the relational schema $R(ABCD)$ with following FD set $F = \{A \rightarrow CE, B \rightarrow D, AE \rightarrow D\}$. Identify the highest normal form satisfied by the relation R.

A. 2NF
B. BCNF
C. 3NF
D. 1NF

nielit2017dec-scientistb databases database-normalization

Answer key☟

6.9.10 Database Normalization: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 16

For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values, only the following functional dependencies and those that can be inferred from them hold.

- $a \rightarrow c$
- $b \rightarrow d$

The relation is in

A. First normal form but not in second normal form
B. Second normal form but not in third normal form
C. Third normal form
D. None of the above

nielit2017oct-assistanta-it databases database-normalization

Answer key☟

6.9.11 Database Normalization: NIELIT Scientific Assistant A 2020 November: 73

Anomalies are avoided by splitting the offending relation into multiple relations, is also known as __________.

A. Accupressure
B. Decomposition
C. Precomposition
D. Both decomposition & precomposition

nielit-sta-2020 databases database-normalization

Answer key☟

6.10 Deadlock Prevention Avoidance Detection (2)

6.10.1 Deadlock Prevention Avoidance Detection: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 34

Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R. It will

A. result in deadlock situation
B. immediately be granted

C. immediately be rejected

D. be granted as soon as it is released by A

nielit2017oct-assistanta-cs databases transaction-and-concurrency deadlock-prevention-avoidance-detection

Answer key☟

6.10.2 Deadlock Prevention Avoidance Detection: NIELIT 2017 OCT Scientific Assistant A (IT) - Section C: 5

Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R. It will

A. result in deadlock situation

B. immediately be granted

C. immediately be rejected

D. be granted as soon as it is released by A

nielit2017oct-assistanta-it databases transaction-and-concurrency deadlock-prevention-avoidance-detection

Answer key☟

6.11 Degree of Graph (1)

6.11.1 Degree of Graph: NIELIT 2021 Dec Scientist B - Section B: 118

Number of entity types participating in $\mathrm{E-R}$ diagrams is represented by :

A. Degree of relationship

B. Structure of relationship

C. Instance of relationship

D. Role of relationship

nielit2021dec-scientistb er-diagram databases degree-of-graph

Answer key☟

6.12 Dependency Preserving (2)

6.12.1 Dependency Preserving: NIELIT 2016 MAR Scientist C - Section C: 3

Every Boyce-Codd Normal Form (BCNF) decomposition is

A. dependency preserving

B. not dependency preserving

C. need be dependency preserving

D. none of these

nielit2016mar-scientistc databases dependency-preserving

Answer key☟

6.12.2 Dependency Preserving: NIELIT 2018-63

Identify the true statement from the given statements.

1. Lossless, dependency-preserving Decompositoin into 3 NF is always possible
2. Any relation with two attributes is BCNF

A. 1

B. 2

C. 1 and 2

D. None of these

nielit-2018 databases dependency-preserving

Answer key☟

6.13 ER Diagram (5)

6.13.1 ER Diagram: NIELIT 2021 Dec Scientist A - Section B: 49

In an ER Diagram, a double ellipse is used to represent :

A. Simple Attribute
B. Composite Attribute
C. Descriptive Attribute
D. Multi-valued Attribute

nielit2021dec-scientista er-diagram databases

Answer key☟

6.13.2 ER Diagram: NIELIT 2021 Dec Scientist B - Section B: 11

In E-R diagrams roles are indicated by labelling the lines that connect which two shapes:

A. Diamond, Diamond
B. Rectangle, Circle
C. Rectangle, Rectangle
D. Diamond, Rectangle

nielit-2021-it-dec-scientistb er-diagram databases

Answer key☟

6.13.3 ER Diagram: NIELIT 2021 Dec Scientist B - Section B: 30

Cardinality of relationship advisor to each entity sets instructor and student will be:

A. One to many
B. One to one
C. Many to many
D. Many to one

nielit-2021-it-dec-scientistb databases er-diagram relations

6.13.4 ER Diagram: NIELIT Scientist B 2020 November: 115

Considering binary relationships, possible cardinality ratios are:

A. one:one
B. $1 : N$
C. $M : N$
D. All the options

nielit-scb-2020 databases er-diagram relations

Answer key☟

6.13.5 ER Diagram: NIELIT Scientist B 2020 November: 43

Set of key attributes that identify weak entities related to some owner entity is classified as:

A. Structural key
B. String key
C. Partial key
D. Foreign key

nielit-scb-2020 databases er-diagram

Answer key☟

6.14 Hashing (1)

6.14.1 Hashing: NIELIT Scientist B 2020 November: 103

The physical location of a record is determined by a mathematical formula that transforms a file key into a record location is:

A. B-Tree File B. Hashed File C. Indexed File D. Sequential File

nielit-scb-2020 databases hashing data-structures

Answer key☟

6.15 Irrecoverable Error (1)

6.15.1 Irrecoverable Error: NIELIT 2022 April Scientist B | Section B | Question: 58

Which of the following scenarios may lead to an irrecoverable error in a database system?

A. A transaction writes a data item after it is read by an uncommitted transaction
B. A transaction reads a data item after it is read by an uncommitted transaction
C. A transaction reads a data item after it is written by a committed transaction
D. A transaction reads a data item after it is written by an uncommitted transaction

nielit2022apr-scientistb databases transaction-and-concurrency irrecoverable-error

6.16 Java (1)

6.16.1 Java: NIELIT 2021 Dec Scientist B - Section B: 66

Which one of the following contains date information?

A. java.sql.TimeStamp B. java.sql.Time
C. java.io.Time D. java.io.TimeStamp

nielit-2021-it-dec-scientistb nielit2021dec-scientistb java databases

6.17 Joins (1)

6.17.1 Joins: NIELIT 2017 July Scientist B (IT) - Section B: 55

A table joined with itself is called

A. Join B. Self Join C. Outer Join D. Equi Join

nielit2017july-scientistb-it databases joins

Answer key☟

6.18 Lock (1)

6.18.1 Lock: NIELIT 2016 MAR Scientist B - Section C: 34

Global locks

A. synchronize access to local resources.
B. synchronize access to global resources.
C. are used to avoid local locks.
D. prevent access to global resources.

nielit2016mar-scientistb databases transaction-and-concurrency lock

Answer key☟

6.19 Lossless Join (1)

6.19.1 Lossless Join: NIELIT 2022 April Scientist B | Section B | Question: 49

Let $\text{R}\;(\text{A}, \text{B}, \text{C}, \text{D})$ be a relational schema with the following functional dependencies:

$\text{A} \rightarrow \text{B}, \;\text{B} \rightarrow \text{C}$

$\text{C} \rightarrow \text{D}$ and $\text{D} \rightarrow \text{B}$

The decomposition of R into

$(\text{A}, \text{B}), (\text{B}, \text{C}), (\text{B}, \text{D})$

A. gives a lossless join, and is dependency preserving
B. gives a lossless join, but is not dependency preserving
C. does not give a lossless join, but is dependency preserving
D. does not give a lossless join and is not dependency preserving

nielit2022apr-scientistb databases database-normalization lossless-join

Answer key☟

6.20 Normal Forms (2)

6.20.1 Normal Forms: NIELIT 2021 Dec Scientist B - Section B: 9

Specify, which of the following is preferred method for enforcing data integrity?

A. Constraints
B. Stored procedure
C. Triggers
D. Cursors

nielit-2021-it-dec-scientistb databases normal-forms database-normalization

Answer key☟

6.20.2 Normal Forms: NIELIT 2022 April Scientist B | Section B | Question: 54

The relation scheme $\text{Student Performance (name, courseNo, rollNo, grade)}$ has the following functional dependencies:

- name, courseNo, $\rightarrow$ grade
- rollNo, courseNo $\rightarrow$ grade
- name $\rightarrow$ rollNo
- rollNo $\rightarrow$ name

The highest normal form of this relation scheme is

A. $2\mathrm{NF}$ B. $3\mathrm{NF}$ C. BCNF D. $4\mathrm{NF}$

nielit2022apr-scientistb databases database-normalization normal-forms functional-dependency

Answer key☟

6.21 Operator Precedence (1)

6.21.1 Operator Precedence: NIELIT 2017 July Scientist B (IT) - Section B: 57

If an SQL query involves NOT,AND,OR with no parenthesis

A. NOT will be evaluated first; AND will be evaluated second; OR will be evaluated last.
B. NOT will be evaluated first; OR will be evaluated second; AND will be evaluated last.
C. AND will be evaluated first; OR will be evaluated second;NOT will be evaluated last.
D. The order of occurrence determines the order of evaluation.

nielit2017july-scientistb-it databases sql operator-precedence

Answer key☟

6.22 Optimization (1)

6.22.1 Optimization: NIELIT 2021 Dec Scientist B - Section B: 56

Which of the following is advantage of using JDBC connection pool?

A. Slow performance
B. Using more memory
C. Using less memory
D. Better performance

nielit-2021-it-dec-scientistb databases optimization

6.23 Primary Key (1)

6.23.1 Primary Key: NIELIT 2021 Dec Scientist A - Section B: 84

An instance of relational schema $\mathrm{R}\,(\mathrm{A}, \mathrm{B}, \mathrm{C})$ has distinct values of A including NULL values. Which one of the following is true ?

A. A is a candidate key
B. A is not a candidate key
C. A is a primary key
D. Both (A) and (C)

nielit2021dec-scientista databases relational-model candidate-key primary-key

Answer key☟

6.24 Query (1)

6.24.1 Query: NIELIT 2022 April Scientist B | Section B | Question: 111

Given the following schema:

employees (emp-id, first-name, last-name, hire-date, dept-id, salary

departments (dept-id, dept-name, manager-id, location-id)

You want to display the last names and hire dates of all latest hires in their respective departments in the location $\text{ID } 1700$. You issue the following query:

```
SQL>SELECT last-name, hire-date
   FROM employees
   WHERE (dept-id, hire-date) IN
   (SELECT dept-id, MAX(hire-date)
   FROM employees JOIN departments USING(dept-id)
   WHERE location-id =1700
   GROUP BY dept-id);
```

What is the outcome?

A. It executes but does not give the correct result
B. It executes and gives the correct result.
C. It generates an error because of pairwise comparison.
D. It generates an error because of the GROUP BY clause cannot be used with table joins in a sub-query.

nielit2022apr-scientistb sql databases query

6.25 Referential Integrity (3)

6.25.1 Referential Integrity: NIELIT 2018-49

The following table has two attributes X and Y where X is the primary key and Y is the foreign key referencing X with on-delete cascade.

X	Y
2	4
3	4
4	3
5	2
7	2
9	5
6	4

The set of all tuples that must be additionally deleted to preserve referential integrity when the tuple $(3, 4)$ is deleted is

A. $(4,3)$ and $(6,4)$ B. $(2,4)$ and $(7,2)$ C. $(3,2)$ and $(9,5)$ D. $(3,4),(4,5)$ and $(6,4)$

nielit-2018 databases referential-integrity

Answer key☟

6.25.2 Referential Integrity: NIELIT 2021 Dec Scientist B - Section B: 10

Mention which of the following is a special type of integrity constraint that relates two relations & maintains consistency across the relations.

A. Entity Integrity Constraints.
B. Referential Integrity Constraints
C. Domain Integrity Constraints
D. Domain Constraints

nielit-2021-it-dec-scientistb relational-model referential-integrity databases

Answer key☟

6.25.3 Referential Integrity: NIELIT Scientific Assistant A 2020 November: 97

Domain constraints, functional dependency and referential integrity are special forms of _______

A. Foreign key
B. Primary key
C. Assertion
D. Referential constraint

nielit-sta-2020 databases database-normalization referential-integrity

Answer key☟

6.26 Relational Algebra (10)

6.26.1 Relational Algebra: NIELIT 2016 DEC Scientist B (IT) - Section B: 18

Which of the following is a fundamental operation in relational algebra?

A. Set intersection
B. Assignment
C. Natural Join
D. None of the above

nielit2016dec-scientistb-it databases relational-algebra

Answer key☟

6.26.2 Relational Algebra: NIELIT 2016 DEC Scientist B (IT) - Section B: 43

Which of the following is a fundamental operation in relational algebra?

A. Set intersection
B. Natural join
C. Assignment
D. None of the above

nielit2016dec-scientistb-it databases relational-algebra

Answer key☟

6.26.3 Relational Algebra: NIELIT 2017 July Scientist B (IT) - Section B: 54

Cross Product is a

A. Unary Operator
B. Ternary Operator
C. Binary Operator
D. Not an operator

nielit2017july-scientistb-it databases relational-algebra

Answer key☟

6.26.4 Relational Algebra: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 27

Which of the following desired features are beyond the capability of relational algebra?

A. Aggregate Computation
B. Multiplication
C. Finding transitive closure
D. All of the above

nielit2017oct-assistanta-it databases relational-algebra

Answer key☟

6.26.5 Relational Algebra: NIELIT 2021 Dec Scientist A - Section B: 72

Consider the relations :

$\mathrm{R_1}\{\mathrm{Roll\_no, Name, Grades}\}$ and $\mathrm{R_2}\{\mathrm{Roll\_no, Subject\_ID, Grades}\}$

Which of the following operations cannot be performed using the above relations?

A. Union
B. Select
C. Join
D. Project

nielit2021dec-scientista relational-algebra databases

Answer key☟

6.26.6 Relational Algebra: NIELIT 2021 Dec Scientist B - Section B: 108

What will be the result of following relational algebra expression ?

$\pi_{\mathrm{A,B}}(\sigma_{\mathrm{C=10}}(\mathrm{R}))$

A. Print columns A & B from relation R when $\mathrm{C} = 10$
B. Print $\mathrm{C} = 10$ from relation R
C. Print all data of relation R when $\mathrm{C} = 10$
D. Print $\mathrm{A, B, C}$ from relation B when $\mathrm{C} = 10$

nielit2021dec-scientistb relational-algebra databases

Answer key☟

6.26.7 Relational Algebra: NIELIT 2021 Dec Scientist B - Section B: 82

The equivalent relational algebra expression for the query “Find the names of suppliers who supplied all the items to all the customers”.

A. $\neg\, t/t \ \in$ supplier $(\mathrm{Q(t)})$
B. $\forall\, t\, [\mathrm{Sname}]\ /\ t \ \in$ supplier $(\mathrm{Q(t)})$
C. $t/\neg\, t \ \in$ supplier $(\mathrm{Q(t)})$
D. $\forall\, t/(\sim \mathrm{Q(t)})$

nielit2021dec-scientistb relational-algebra databases

6.26.8 Relational Algebra: NIELIT 2022 April Scientist B | Section B | Question: 66

Given the relations

employee (name, salary, deptno) and

department (deptno, deptname, address)

Which of the following queries cannot be expressed using the basic relational algebra operations $(\mathrm{U}, -, \mathrm{x}, \ , \ , \mathrm{p})$?

A. Department address of every employee
B. Employees whose name is the same as their department name
C. The sum of all employees' salaries
D. All employees of a given department

nielit2022apr-scientistb relational-algebra databases

6.26.9 Relational Algebra: NIELIT Scientific Assistant A 2020 November: 119

Let R and S be two relations with the following schema

$R(\underline{P, Q,}R1, R2, R3)$

$S(\underline{P, Q,}S1, S2)$

Where [P,Q] is the key for both schemas. Which of the following queries are equivalent?

(I) $II_P(R \bowtie S)$

(II) $II_P(R) \bowtie II_P(S)$

(III) $II_P(II_{P,Q}(R) - (II_{P,Q}(R) - II_{P,Q}(S)))$

A. Only (I) and (II)
B. Only (I) and (III)
C. Only (I), (II) and (III)
D. Only (I), (III) and (IV)

nielit-sta-2020 databases relational-algebra

Answer key☟

6.26.10 Relational Algebra: NIELIT Scientific Assistant A 2020 November: 48

What is meant by the following relational algebra statement: $\mathrm{STUDENT} \times \mathrm{COURSE}$?

A. Compute the natural join between the $\mathrm{STUDENT}$ and COURSE relations
B. Compute the left outer join between the $\mathrm{STUDENT}$ and COURSE relations
C. Compute the cartesian product between the $\mathrm{STUDENT}$ and COURSE relations
D. Compute the outer join between the $\mathrm{STUDENT}$ and COURSE relations

nielit-sta-2020 databases relational-algebra

Answer key☟

6.27 Relational Calculus (4)

6.27.1 Relational Calculus: NIELIT 2016 MAR Scientist B - Section C: 42

The relational algebra expression equivalent to the tuple calculus expression

$\{t \mid t \in r \wedge (t[A] = 10 \wedge t[B] = 20)\}$ is

A. $\sigma_{(A=10 \vee B=20)}(r)$
B. $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$
C. $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$
D. $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

nielit2016mar-scientistb databases relational-calculus

Answer key

6.27.2 Relational Calculus: NIELIT 2017 July Scientist B (IT) - Section B: 19

If R is a relation in Relational Data Model and $A_1, A_2, \ldots A_n$ are the attributes of relation R, what is the cardinality of R expressed in terms of domain of attributes?

A. $|R| \leq |\text{ dom }(A_1) \times \text{ dom }(A_2) \ldots \text{ dom }(A_n)|$
B. $|R| \geq |\text{ dom } \quad (A_1) \times \text{ dom } \quad (A_2) \ldots \text{ dom } \quad (A_n)|$
C. $|R| = \text{ max }(|\text{ dom}(A_1)|, |\text{ dom }(A_2)|, \cdots |\text{ dom }(A_n)|)$
D. $|R| = \text{ min }(|\text{ dom}(A_1)|, |\text{ dom }(A_2)|, \cdots |\text{ dom }(A_n)|)$

nielit2017july-scientistb-it databases relational-model relational-calculus

Answer key

6.27.3 Relational Calculus: NIELIT 2022 April Scientist B | Section B | Question: 83

Which of the following relational calculus expressions is not safe ?

i. $\{t \mid \exists u \in R_1 (t[A] = u[A]) \wedge \neg \exists s \in R_2 (t[A] = s[A])\}$
ii. $\{t \mid \forall u \in R_1 (u[A] = " x " \Rightarrow \exists s \in R_2 (t[A] = s[A] \wedge s[A] = u[A]))\}$
iii. $\{t \mid \neg (t \in R_1)\}$
iv. $\{t \mid \exists u \in R_1 (t[A] = u[A]) \wedge \exists s \in R_2 (t[A] = s[A])\}$

A. (i)
B. (ii)
C. (iii)
D. (iv)

nielit2022apr-scientistb relational-calculus databases

6.27.4 Relational Calculus: NIELIT Scientist B 2020 November: 111

An expression in the domain relational calculus is of the form:

A. $\{P(x_1, x_2, \ldots, x_n) | < x_1, x_2, \ldots, x_n >\}$
B. $\{x_1, x_2, \ldots, x_n | < x_1, x_2, \ldots, x_n >\}$
C. $\{x_1, x_2, \ldots, x_n | x_1, x_2, \ldots, x_n\}$
D. $\{< x_1, x_2, \ldots, x_n > | P(x_1, x_2, \ldots, x_n)\}$

nielit-scb-2020 databases relational-calculus

Answer key

6.28 Relational Model (11)

6.28.1 Relational Model: NIELIT 2016 MAR Scientist B - Section C: 41

The employee salary should not be greater than Rs.2000. This is

A. integrity constraint
B. referential constraint
C. over-defined constraint
D. feasible constraint

nielit2016mar-scientistb databases relational-model

Answer key

6.28.2 Relational Model: NIELIT 2016 MAR Scientist C - Section C: 5

A primary key, if combined with a foreign key creates

A. parent child relationship between the tables that connect them
B. many-to-many relationship between the tables that connect them
C. network model between the tables that connect them
D. none of these

nielit2016mar-scientistc databases relational-model

Answer key

6.28.3 Relational Model: NIELIT 2017 DEC Scientist B - Section B: 34

The condition for total participation of entity in a relationship is _______.

A. Maximum cardinality should be one
B. Minimum cardinality should be one
C. Minimum cardinality should be zero
D. None of the options

nielit2017dec-scientistb databases relational-model

Answer key

6.28.4 Relational Model: NIELIT 2017 July Scientist B (IT) - Section B: 53

Related fields in a database are grouped to form a

A. data file
B. data record
C. menu
D. bank

nielit2017july-scientistb-it databases relational-model

Answer key

6.28.5 Relational Model: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 14

E-R model uses this symbol to represent weak entity set?

A. Dotted rectangle
B. Diamond
C. Doubly outlined rectangle
D. None of these

nielit2017oct-assistanta-cs databases relational-model

Answer key

6.28.6 Relational Model: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 21

What is the modality of relationship, if there is no explicit need for relationship to occur?

A. Zero B. Two C. Three D. One

nielit2017oct-assistanta-cs databases relational-model

Answer key

6.28.7 Relational Model: NIELIT 2018-42

_____ symbol is used to denote derived attributes in ER model

A. Dashed ellipse
B. Squared ellipse
C. Ellipse with attribute name underlined
D. Rectangular Box

nielit-2018 databases relational-model

Answer key

6.28.8 Relational Model: NIELIT 2021 Dec Scientist B - Section B: 51

The ____________ of a relationship is 0 if there is no explicit need for the relationship to occur or the relationship is optional.

A. modality B. cardinality C. entity D. structured analysis

nielit2021dec-scientistb databases er-diagram relational-model

Answer key

6.28.9 Relational Model: NIELIT 2021 Dec Scientist B - Section B: 58

Indicate which representation is the logical design of the database, and which is a snapshot of the data in the database at a given instant in time.

A. Instance, Range
B. Relation, Schema
C. Relation, Domain
D. Schema, Instance

nielit-2021-it-dec-scientistb databases relational-model

Answer key

6.28.10 Relational Model: NIELIT 2021 Dec Scientist B - Section B: 72

Consider a statement

```
Course (course_id, sec_id, semester)
```

Here the $\text{course\_id, sec\_id, and semester}$ will be termed as _______ and Course is a ________.

A. Relations, Attribute
B. Attributes, Relation
C. Tuple, Relation
D. Tuple, Attribute

nielit-2021-it-dec-scientistb relational-model databases

Answer key☟

6.28.11 Relational Model: NIELIT Scientist B 2020 November: 87

Point out the wrong statement :

A. Non-Relational databases require that schemas be defined before you can add data
B. No SQL databases are built to allow the insertion of data without a predefined schema
C. New SQL databases are built to allow the insertion of data without a predefined schema
D. All of the options

nielit-scb-2020 databases relational-model

Answer key☟

6.29 SQL (9)

6.29.1 SQL: NIELIT 2016 DEC Scientist B (IT) - Section B: 41

Which type of Statement can execute parameterized queries?

A. PreparedStatement
B. ParameterizedStatement
C. ParameterizedStatement and CallableStatement
D. All kinds of Statements

nielit2016dec-scientistb-it databases sql

Answer key☟

6.29.2 SQL: NIELIT 2017 July Scientist B (IT) - Section B: 52

'AS' clause is used in SQL for

A. Selection operation
B. Rename operation
C. Join Operation
D. Projection Operation

nielit2017july-scientistb-it databases sql

Answer key☟

6.29.3 SQL: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 13

Given relations $R(w, x)$ and $S(y, z)$, the result of
SELECT DISTINCT w, x from R, S

A. R has no duplicates and S is non-empty
B. R and S have no duplicates
C. S has no duplicates and R is non-empty
D. R and S has the same number of tuples

nielit2017oct-assistanta-cs databases sql

Answer key☟

6.29.4 SQL: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 36

Table employees has 10 records. It has a non-NULL SALARY column which is also UNIQUE. The SQL statement

SELECT COUNT(*)

FROM EMPLOYEE

WHERE SALARY > ALL (SELECT SALARY FROM EMPLOYEE);

A. 10 B. 9 C. 5 D. 0

nielit2017oct-assistanta-cs databases sql

Answer key

6.29.5 SQL: NIELIT 2021 Dec Scientist B - Section B: 74

DELETE [FROM] table $[\text{ WHERE condition}];$ from the syntax if you omit the WHERE clause.

A. All rows in the table are deleted.
B. It will give you an error.
C. No rows will be deleted.
D. Only one row will be deleted.

nielit2021dec-scientistb sql databases

Answer key

6.29.6 SQL: NIELIT Scientific Assistant A 2020 November: 45

$(< \text{ALL})$ comparison operator means:

A. more than the maximum value in the subquery
B. less than the minimum value in the subquery
C. is equivalent to IN
D. none of the options

nielit-sta-2020 databases sql

Answer key

6.29.7 SQL: NIELIT Scientific Assistant A 2020 November: 91

With the following syntax

```
INSERT INTO table[(column[,column...])]
VALUES (value[, value...]);
```

you can:

A. Insert one row at a time
B. Insert multiple rows at a time
C. Insert one column at a time
D. Insert multiple columns at a time

nielit-sta-2020 databases sql

Answer key

6.29.8 SQL: NIELIT Scientific Assistant A 2020 November: 98

When retrieving data in particular table in PostgreSQL, we use the _______ statement.

A. \dt B. ORDER BY C. SELECT FROM D. \i

nielit-sta-2020 databases sql

Answer key

6.29.9 SQL: NIELIT Scientist B 2020 November: 99

Limitations of the XML Data Type are:

A. It cannot be compared or sorted. This means an XML data type cannot be used in a GROUP BY statement
B. It cannot be used as a key column in an index
C. The value() method of the XML data type returns a scalar value, so it can be specified anywhere where scalar values are allowed
D. All of the options

nielit-scb-2020 databases sql database-normalization

Answer key

6.30 Serializability (3)

6.30.1 Serializability: NIELIT 2017 July Scientist B (CS) - Section B: 46

Consider the following four schedules due to three transactions (indicated by the subscript) using read and write on a data item x, denoted by $r(x)$ and $w(x)$respectively. Which one of them is conflict serializable?

1. $r_1(x); r_2(x); w_1(x); r_3(x); w_2(x)$
2. $r_2(x); r_1(x); w_2(x); r_3(x); w_1(x)$
3. $r_3(x); r_2(x); r_1(x); w_2(x); w_1(x)$
4. $r_2(x); w_2(x); r_3(x); r_1(x); w_1(x)$

A. 1 B. 2 C. 3 D. 4

nielit2017july-scientistb-cs databases serializability conflict-serializable

6.30.2 Serializability: NIELIT 2021 Dec Scientist B - Section B: 107

If all transactions obey the ______________ then all possible interleaved schedules (non-serial schedules) are serializable.

A. Lock based protocol B. Two phase Locking protocal
C. Read-write lock based protocol D. Time stamp based protocol

nielit2021dec-scientistb transaction-and-concurrency serializability databases

6.30.3 Serializability: NIELIT 2022 April Scientist B | Section B | Question: 57

Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{\mathrm{O1}, \ldots, \mathrm{Ok}\}$. This is done in the following manner:

- $\mathrm{Step}\ 1.\ \mathrm{T}$ acquires exclusive locks to $\mathrm{O1}, \ldots, \mathrm{Ok}$ in increasing order of their addresses.
- $\mathrm{Step}\ 2.$ The required operations are performed.
- $\mathrm{Step}\ 3.$ All locks are released

This protocol will

A. guarantee serializability and deadlock-freedom
B. guarantee neither serializability nor deadlock-freedom
C. guarantee serializability but not deadlock-freedom
D. guarantee deadlock-freedom but not serializability.

nielit2022apr-scientistb transaction-and-concurrency deadlock-prevention-avoidance-detection serializability databases

Answer key☟

6.31 Super Key (1)

6.31.1 Super Key: NIELIT 2018-55

Maximum number of superkeys for the relation schema $R(X, Y, Z, W)$ with X as the key is

A. 6 B. 8 C. 9 D. 12

nielit-2018 databases super-key

Answer key☟

6.32 Transaction and Concurrency (5)

6.32.1 Transaction and Concurrency: NIELIT 2017 DEC Scientist B - Section B: 18

In conservative two phase locking protocol, a transaction

A. Should release all the locks only at the beginning of transaction
B. Should release exclusive locks only after the commit operation
C. Should acquire all the exclusive locks at the beginning of transaction
D. Should acquire all the locks at the beginning of transaction

nielit2017dec-scientistb databases transaction-and-concurrency two-phase-locking-protocol

Answer key☟

6.32.2 Transaction and Concurrency: NIELIT 2017 OCT Scientific Assistant A (CS) - Section C: 10

When transaction Ti requests a data item currently held by Tj, Ti is allowed to wait only if it has a timestamp smaller than that of Tj (that is Ti is order than Tj). Otherwise, Ti is rolled back (dies). This is

A. Wait-die B. Wait-wound C. Wound-wait D. Wait

nielit2017oct-assistanta-cs databases transaction-and-concurrency

Answer key

6.32.3 Transaction and Concurrency: NIELIT 2017 OCT Scientific Assistant A (IT) - Section B: 31

When transaction Ti requests a data item currently held by Tj, Ti is allowed to wait only if it has a time stamp smaller than that of Tj (that is Ti is order than Tj). Otherwise, Ti is rolled back (dies). This is

A. Wait-die B. Wait-wound C. Wound-wait D. Wait

nielit2017oct-assistanta-it databases transaction-and-concurrency

Answer key

6.32.4 Transaction and Concurrency: NIELIT 2021 Dec Scientist B - Section B: 78

Which of the following is not a JDBC connection isolation levels?

A. TRANSACTION_NONE
B. TRANSACTION_READ_COMMITTED
C. TRANSACTION_REPEATABLE_READ
D. TRANSACTION_NONREPEATABLE_R

nielit-2021-it-dec-scientistb databases transaction-and-concurrency sql

6.32.5 Transaction and Concurrency: NIELIT Scientific Assistant A 2020 November: 78

Which of the following scenarios may lead to an irrecoverable error in a database system?

A. A transaction writes a data item after it is read by an uncommitted transaction
B. A transaction reads a data item after it is read by an uncommitted transaction
C. A transaction reads a data item after it is written by a committed transaction
D. A transaction reads a data item after it is written by an uncommitted transaction

nielit-sta-2020 databases transaction-and-concurrency

Answer key

6.33 Transactions and Concurrency Control (1)

6.33.1 Transactions and Concurrency Control: NIELIT 2018-36

______ is NOT a part of the ACID properties

A. Inconsistency B. Consistency C. Atomicity D. Isolation

nielit-2018 databases transactions-and-concurrency-control

Answer key

6.34 Tuple Relational Calculus (3)

6.34.1 Tuple Relational Calculus: NIELIT 2016 MAR Scientist B - Section C: 40

In a relational Schema, each tuple is divided into fields called

A. relations B. domains C. queries D. none of these

nielit2016mar-scientistb databases relational-model tuple-relational-calculus

Answer key☟

6.34.2 Tuple Relational Calculus: NIELIT 2017 July Scientist B (IT) - Section B: 56

Consider the join of a relation R with relation S. If R has m tuples and S has n tuples, then the maximum size of join is

A. mn B. $m+n$ C. $(m+n)/2$ D. $2(m+n)$

nielit2017july-scientistb-it databases joins tuple-relational-calculus

Answer key☟

6.34.3 Tuple Relational Calculus: NIELIT 2017 July Scientist B (IT) - Section B: 58

In tuple relational calculus $P1 \rightarrow P2$ is equivalent to

A. $\neg P1 \vee P2$
B. $P1 \vee P2$
C. $P1 \wedge P2$
D. $P1 \wedge \neg P2$

nielit2017july-scientistb-it databases tuple-relational-calculus

Answer key☟

6.35 Undo Redo (1)

6.35.1 Undo Redo: NIELIT Scientist B 2020 November: 80

Consider a Simple Checkpointing Protocol and the following set of operations in the log.

$(start, T_4); (write, T_4, y, 2, 3); (start, T_1);$
$(commit, T_4); (write, T_1, z, 5, 7);$
$(Checkpoint);$
$(start, T_2); (write, T_2, x, 1, 9);$
$(commit, T_2); (start, T_3); (write, T_3, z, 7, 2);$

if a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list?

A. $Undo: T_3, T_1; Redo\ T_2$
B. $Undo: T_3, T_1; Redo\ T_2, T_4$
C. $Undo: none;\ Redo\ T_2, T_4, T_3, T_1$
D. $Undo: T_3, T_1, T_4;\ Redo\ T_2$

nielit-scb-2020 transaction-and-concurrency databases undo-redo checkpointing

Answer key

6.36 Unique Key (1)

6.36.1 Unique Key: NIELIT 2021 Dec Scientist A - Section B: 83

Consider the following statements :

I. The primary key of a relation cannot contain null values.
II. Unique Key can have null values.

Which among the following is true ?

A. Both I and II are true
B. Both I and II are false
C. Only I is true
D. Only II is true

nielit2021dec-scientista databases relational-model primary-key unique-key

Answer key

6.37 Web Technologies (1)

6.37.1 Web Technologies: NIELIT Scientist B 2020 November: 68

How to specify the comment in the XML document?

A. <?-- -->
B. <!-- --!>
C. <!-- -->
D. </-- -->

nielit-scb-2020 web-technologies

Answer key

Answer Keys

6.0.1	B	6.0.2	D	6.0.3	C	6.1.1	D	6.2.1	A
6.2.2	D	6.2.3	A	6.3.1	C	6.3.2	A	6.4.1	B
6.4.2	D	6.5.1	B	6.6.1	C	6.6.2	C	6.6.3	B
6.6.4	B	6.6.5	B	6.7.1	B	6.7.2	D	6.7.3	A
6.7.4	D	6.8.1	A	6.8.2	C	6.9.1	A	6.9.2	C
6.9.3	B	6.9.4	A	6.9.5	D	6.9.6	C	6.9.7	D
6.9.8	A	6.9.9	D	6.9.10	A	6.9.11	B	6.10.1	B
6.10.2	B	6.11.1	A	6.12.1	D	6.12.2	C	6.13.1	D
6.13.2	D	6.13.3	A	6.13.4	D	6.13.5	C	6.14.1	B
6.15.1	D	6.16.1	A	6.17.1	B	6.18.1	B	6.19.1	A
6.20.1	A	6.20.2	B	6.21.1	A	6.22.1	D	6.23.1	A

6.24.1	B
6.26.2	D
6.26.7	B
6.27.2	A
6.28.3	B
6.28.8	A
6.29.2	B
6.29.7	A
6.30.3	A
6.32.4	D
6.34.3	A

6.25.1	X
6.26.3	C
6.26.8	C
6.27.3	C
6.28.4	B
6.28.9	D
6.29.3	A
6.29.8	C
6.31.1	B
6.32.5	D
6.35.1	A

6.25.2	B
6.26.4	D
6.26.9	D
6.27.4	D
6.28.5	C
6.28.10	B
6.29.4	D
6.29.9	D
6.32.1	D
6.33.1	A
6.36.1	A

6.25.3	C
6.26.5	A
6.26.10	C
6.28.1	A
6.28.6	A
6.28.11	A
6.29.5	A
6.30.1	D
6.32.2	A
6.34.1	B
6.37.1	C

6.26.1	D
6.26.6	A
6.27.1	C
6.28.2	A
6.28.7	A
6.29.1	A
6.29.6	B
6.30.2	B
6.32.3	A
6.34.2	A